Real regularity of complex Hessian obstacle envelopes

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Abstract

The complex m-Hessian obstacle envelope is locally $C^{1,1}$ in all real coordinates for every $1 \le m \le n$, under the stated strict defining-function assumption. The penalized Bellman estimates are uniform in both approximation parameters. The conclusion concerns exactly the catalogue target; no historical priority claim is made.

Catalogue scope. AIM 721; complete resolution. Canonical target at revision `1e56463`. The domain-envelope question is asked in [1, p. 998]. The accompanying erratum [2] is retained in the references; its incomplete original boundary-data argument is not used here.

1 Statement and argument

The exact obstacle-envelope question has an affirmative answer under the stated strict defining-function hypothesis. The argument uses smooth Bellman approximants with a quadratic stopping penalty. The essential new estimate compensates the discount when the diffusion clock is perturbed; this keeps the large discount out of the second-variation bound. We give the approximation, control comparison, and limiting argument explicitly.

Theorem 1.1. *Let $\Omega \Subset \mathbb{C}^n$ have smooth boundary and a smooth defining function ρ on a neighborhood of $\overline{\Omega}$ such that $\rho < 0$ in Ω, $d\rho \neq 0$ on $\partial\Omega$, and ρ is strictly m-subharmonic on that neighborhood. If $h \in C^2(\overline{\Omega}; \mathbb{R})$, then*

$$P_{m,\Omega} h := \big(\sup\{w \in \mathrm{SH}_m(\Omega) : w \le h\}\big)^* \in C^{1,1}_{\mathrm{loc}}(\Omega) \cap C^{0,1}(\overline{\Omega}). \tag{1}$$

Here $C^{1,1}$ concerns the full real Hessian. The conclusion holds for all $1 \le m \le n$, including the intermediate orders in AIM 721.

1.1 Covariances and smooth approximants

Let $\Gamma_m \subset \mathrm{Herm}(n)$ be the open Gårding cone, and put

$$\mathcal{A} = \{A \in \overline{\Gamma}_m^* : \operatorname{tr} A = 1\}, \qquad L_A u = \operatorname{tr}(A u_{z\bar{z}}). \tag{2}$$

The cone contains the positive definite cone, so its dual consists of positive semidefinite matrices. Consequently $\mathcal{A}$ is compact and convex. It is invariant under unitary conjugation and contains I/n, since $\Gamma_m \subset \{H : \operatorname{tr} H > 0\}$. The constant-coefficient dual-cone characterization says that u is m-subharmonic if and only if $L_A u \ge 0$ for every $A \in \mathcal{A}$, in the distributional, equivalently viscosity,

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

sense. For smooth functions this is convex duality; the general statement follows by convolution and the usual viscosity characterization of constant-coefficient linear subharmonicity.

Strictness and compactness give $L_A\rho \geq \delta_0 > 0$ uniformly on $\overline{\Omega}$ and $A \in \mathcal{A}$. Scale $-\rho$ to obtain ψ with

$$\Omega = \{\psi > 0\}, \qquad \psi = 0 \text{ on } \partial\Omega, \qquad |\nabla\psi| \geq 1 \text{ on } \partial\Omega, \qquad L_A\psi \leq -1. \tag{3}$$

Enlarge this fixed scaling, if necessary, so that $-L_A\psi \geq 4\sup_A \|\sigma_A\|^2$, where in real coordinates $L_A = (1/2)\sigma_A\sigma_A^{\mathsf{T}} : D^2$. This is the normalization used immediately before [3, Lemma 7.1]. All covariance traces are bounded uniformly.

Set $g = -h$ and, for $0 < \nu < 1$, define

$$\mathcal{A}_\nu = \{(1-\nu)A + \nu I/n : A \in \mathcal{A}\}, \qquad F_\nu(D^2v) = \sup_{A\in\mathcal{A}_\nu} L_A v. \tag{4}$$

The sets $\mathcal{A}_\nu$ are unitary invariant, contain I/n, and satisfy (3). Each fixed ν gives a uniformly elliptic operator. The estimates below use an upper bound for the covariances but no positive lower bound. Initially take g smooth on a neighborhood of $\overline{\Omega}$.

Choose

$$C_0 \geq \sup_{\nu, A\in\mathcal{A}_\nu} \|L_A g\|_\infty, \qquad C_* = C_0 + 1, \qquad b_K = \sqrt{2C_*K}, \qquad K \geq 1,$$

and write

$$H_b(s) = \max_{0\leq c\leq b}\left(cs - \frac{c^2}{2K}\right).$$

For fixed K, ν solve

$$F_\nu(D^2v) + H_{2b_K}(g - v) = 0 \quad \text{in } \Omega, \qquad v = g \quad \text{on } \partial\Omega. \tag{5}$$

This is a uniformly elliptic Bellman equation with compact controls $(A, c) \in \mathcal{A}_\nu \times [0, 2b_K]$ and running payoff $cg - c^2/(2K)$. Standard nondegenerate Bellman Dirichlet theory gives $v = v_{K,\nu} \in C^{2,\alpha}(\overline{\Omega})$ for some $\alpha > 0$. Precisely this fixed-parameter regularization is used in [3, §5, Remark 5.1], citing [4, §6.2, Theorem 7]. Only the existence of a C^2 solution for fixed parameters is used here, not a uniform estimate from that theorem.

The two discount caps. At a positive maximum of $w = g - v$, one has $D^2v \geq D^2g$, whence

$$H_{2b_K}(w) = -F_\nu(D^2v) \leq -F_\nu(D^2g) \leq C_0.$$

If $w \geq b_K/K$, the admissible choice $c = b_K$ would give $H_{2b_K}(w) \geq C_* > C_0$. Thus $w < b_K/K$ everywhere, the maximizing discount is $c = K(g - v)_+$, and

$$F_\nu(D^2v) + \frac{K}{2}(g - v)_+^2 = 0, \qquad (g - v)_+ \leq \sqrt{2C_0/K}. \tag{6}$$

The same v therefore solves the Bellman problem with the smaller cap $[0, b_K]$, and verification identifies it with the value function for either cap. We shall use base controls $c_t \leq b_K$, so that

$$q_t := \frac{c_t^2}{2K} \leq C_*. \tag{7}$$

A perturbed control will use the larger cap. This bound holds for every base control, without any assumption of exact optimality.

Uniform value and gradient bounds. For a fixed sufficiently large C, $g + C\psi$ and $g - C\psi$ are respectively a supersolution and a subsolution of (5). Indeed the first has $F_\nu(D^2(g + C\psi)) \le C_0 - C$ and zero penalty, whereas the second has $F_\nu(D^2(g - C\psi)) \ge -C_0 + C$ and nonnegative penalty. Thus

$$|v - g| \le C\psi. \tag{8}$$

For a small real translation a, put $v_a(x) = v(x + a) - L|a|$ on $\Omega \cap (\Omega - a)$. If $L \ge \mathrm{Lip}(g)$, then $g(x) - v_a(x) \ge g(x + a) - v(x + a)$, so v_a is a subsolution of (5). On the boundary of the intersection, one of $x, x + a$ lies on $\partial\Omega$. Formula (8) gives $v_a \le v$ there if $L \ge \mathrm{Lip}(g) + C\,\mathrm{Lip}(\psi)$. Comparison, followed by the reversed translation, proves

$$\|v\|_{C^{0,1}(\overline{\Omega})} \le C \tag{9}$$

uniformly in K, ν. The Lipschitz constants of the fixed smooth extensions of g, ψ are finite. Larger translation lengths are harmless by the uniform value bound. In particular the boundary normal derivative is uniformly bounded.

1.2 The covariance quasiderivatives and their estimates

We use the covariance constructions in [3, Lemmas 6.2–6.4 and 7.4]; theorem and equation numbers here refer to the inspected arXiv version. These constructions concern the controlled diffusion and its defining-function barrier and have no running payoff or discount. The unitary adaptation requires a sign convention and one drift term to be made explicit.

Near $\partial\Omega$, let η be the complex column vector corresponding to $\nabla\psi$, and let η_ξ denote its real directional derivative. Set

$$r_0 = -\frac{\mathrm{Re}(\eta^*\eta_\xi)}{|\eta|^2}, \qquad b = -\eta_\xi - r_0\eta, \qquad Q = \frac{b\eta^* - \eta b^*}{|\eta|^2} - \frac{\eta^* b}{|\eta|^4}\eta\eta^*. \tag{10}$$

Since $\mathrm{Re}(\eta^* b) = 0$, Q is skew Hermitian, $Q\eta = b$, and $|Q| + |r_0| \le C|\xi|$. Set $P = -\,\mathrm{Real}(Q)$, the negative of its real representation. Then P is skew symmetric, its exponential represents a unitary map, and

$$D^2\psi\,\xi + r_0\nabla\psi - P\nabla\psi = 0, \qquad \psi_{\xi\sigma^k} + r_0\psi_{\sigma^k} + \psi_{P\sigma^k} = 0. \tag{11}$$

The minus sign in P is required by the column-vector convention. This supplies the rotation allowed by the covariance cone; full orthogonal invariance in real dimension $2n$ is not needed. Compare the complex construction in [5, §3.2, equations (3.14)–(3.15)].

There is a minor correction to the source calculation that is useful to record. The displayed assertion in the proof of [3, Lemma 7.1] that

$$S := \sum_k (D^2\psi\,\sigma^k, P\sigma^k) = \mathrm{tr}(\sigma\sigma^{\mathsf{T}} D^2\psi P) = 0$$

is not true in general. Instead $|S| \le C|\xi|$. Keeping this drift in that proof changes its term J_4 to

$$J_4 = 2\frac{\psi_\xi}{\psi}((L\psi)_\xi + 2r_0 L\psi + S) \le C|\xi|^2 + \frac{|\psi_\xi|^2}{\psi^2}. \tag{12}$$

This is the same bound used there, with an enlarged geometric constant. Taking the fixed geometric parameter $\lambda_0 > 0$ sufficiently small therefore gives the same negative drift for its barrier B_1. The remaining first-quasiderivative estimates use (11), skew symmetry, $|P| + |r_0| \le C|\xi|$, and this negative drift; derivatives of P are not needed. Thus the unitary version has those estimates as well.

Here is the precise input we need. Let $D_\delta = \{\psi > \delta\}$ and let X_t be a base controlled diffusion, starting at $x \in D_\delta$, with exit time τ_δ. For a real vector ξ, the constructions supply first and second quasiderivatives ξ_t, η_t, with $\xi_0 = \xi$, $\eta_0 = 0$, scalar adjoint process Z_t, and adjustments $r_t, \widehat{r}_t, \pi_t, \widehat{\pi}_t, P_t$. In the rest of the proof η_t denotes this second quasiderivative, independently of the gradient vector used in (10). We have

$$dZ_t = \pi_t\,dW_t, \qquad Z_0 = 0, \qquad \widehat{\pi}_t = -2\pi_t Z_t, \qquad \widehat{P}_t = 0.$$

The perturbed states X_t^s start at $x + s\xi$ and have derivatives ξ_t, η_t at $s = 0$. Their density $p_t(s)$ satisfies

$$p_t(0) = 1, \qquad p_t'(0) = Z_t, \qquad p_t''(0) = 0. \tag{13}$$

The last identity follows from the choice of $\widehat{\pi}$ and Itô's formula for Z_t^2. Their time factor is

$$\theta_t(s) = 1 + \pi^{-1} \arctan\Big(\pi(2sr_t + s^2\widehat{r}_t)\Big), \quad \tfrac{1}{2} < \theta_t(s) < \tfrac{3}{2}, \quad \theta_t'(0) = 2r_t, \quad \theta_t''(0) = 2\widehat{r}_t. \tag{14}$$

There is a positive quadratic barrier $B(x, \xi)$, bounded by $C_E|\xi|^2$ when x ranges over any compact $E \Subset \Omega$, such that

$$\mathbb{E} \sup_{t \le \tau_\delta} (|\xi_t|^2 + |Z_t|^2) + \mathbb{E} \int_0^{\tau_\delta} \left(|\xi_t|^2 + \frac{|\psi_{\xi_t}(X_t)|^2}{\psi(X_t)^2} + r_t^2 + |\widehat{r}_t| + |\pi_t|^2 \right) dt \le CB(x, \xi). \tag{15}$$

These are [3, Lemma 7.4(1)–(3)] and the explicit coefficient formulas. Near the boundary $r = r_0 + \psi_\xi/\psi$ and $\widehat{r} = (\psi_\xi/\psi)^2$; in the interior $\psi \ge \lambda_0^2$. For fixed δ there is also $\ell = \ell(\delta) > 0$ such that

$$\sup_{\text{controls}} \sup_{\gamma \le \tau_\delta} \mathbb{E}|\xi_\gamma|^{2(1+\ell)} < \infty, \tag{16}$$

by [3, Lemma 7.4(4)]. All constants in (15) depend only on the domain, barrier, dimension and upper covariance bound. In particular they are independent of K, ν, δ.

An additional occupation estimate and a direct second-derivative estimate. We shall also need

$$\mathbb{E} \int_0^{\tau_\delta} Z_t^2 dt \le CB(x, \xi). \tag{17}$$

Apply Itô's formula, with localization, to $\psi(X_t)Z_t^2$. Using $L_A\psi \le -1$ and bounded covariances and gradient of ψ, its drift is at most

$$-Z_t^2 + C|\pi_t|^2 + 2C|Z_t\pi_t| \le -\tfrac{1}{2}Z_t^2 + C'|\pi_t|^2.$$

The initial product is zero and the terminal product is nonnegative. Expectation, followed by (15) and monotone convergence, proves (17). It follows that

$$\mathbb{E} \int_0^{\tau_\delta} (|r_t Z_t| + |\pi_t Z_t|) dt \le CB(x, \xi). \tag{18}$$

This is Cauchy–Schwarz on the product of probability and time measures.

The estimate for η_t can now be proved directly, since the underlying diffusion has zero drift. By the source's equation (6.4),

$$\begin{aligned} d\eta_t &= G_t\, dW_t + H_t\, dt, \\ G_t &= ((\widehat{r}_t - r_t^2)I + P_t^2 + 2r_t P_t)\sigma_t, \\ H_t &= -\sigma_t\widehat{\pi}_t - 2(r_t I + P_t)\sigma_t\pi_t. \end{aligned}$$

Put $w_t = \psi_{\xi_t}(X_t)/\psi(X_t)$. Near the boundary the w_t^2 terms in $\widehat{r}_t - r_t^2$ cancel. In the interior, $\psi \ge \lambda_0^2$ implies $w_t = O_{\lambda_0}(|\xi_t|)$. Hence globally

$$|G_t| \le C|\xi_t|(|\xi_t| + |w_t|), \qquad |H_t| \le C(|\xi_t|^2 + |w_t|^2) + C|\pi_t Z_t|.$$

The Burkholder–Davis–Gundy inequality and (15)–(18) give

$$\begin{aligned} \mathbb{E} \sup_{t \le \tau_\delta} |\eta_t| &\le C\mathbb{E} \left(\int_0^{\tau_\delta} |G_t|^2 dt \right)^{1/2} + \mathbb{E} \int_0^{\tau_\delta} |H_t| dt \\ &\le C \left(\mathbb{E} \sup |\xi_t|^2 \right)^{1/2} \left(\mathbb{E} \int (|\xi_t|^2 + |w_t|^2) dt \right)^{1/2} + CB(x, \xi) \le CB(x, \xi). \end{aligned} \tag{19}$$

Thus no extra integrability assertion for a random exit time times a supremum is required.

The unchanged terminal comparison. Near the boundary, the source's barrier is

$$B_1(y,\zeta) = \gamma(y)\left(\beta(y)|\zeta|^2 + \frac{|\psi_\zeta(y)|^2}{\psi(y)}\right), \qquad \gamma(y) \longrightarrow \lambda_0^2 > 0, \quad \beta(y) \longrightarrow 1. \tag{20}$$

Let γ_s be the minimum of the exit times of X, X^s, X^{-s} from D_δ, a bounded time localization, and quasiderivative localizations. For a fixed C^2 function v, the terminal transfer in [3, equation (9.16)] gives, after taking $s \to 0$ and removing the other localizations,

$$\limsup \sup_{\text{controls}} \mathbb{E}[-e^{-\phi_{\gamma_s}} v_{\xi_{\gamma_s}\xi_{\gamma_s}}(X_{\gamma_s})] \le CB(x,\xi)\left(1 + \sup_{\psi(y)=\delta,\,\zeta\neq 0} \frac{|v_{\zeta\zeta}(y)|}{B_1(y,\zeta)}\right). \tag{21}$$

This holds for any nonnegative accumulated discount ϕ: the terminal comparison uses only $e^{-\phi} \le 1$, and has no derivatives of ϕ. We spell out the needed uniform-integrability point below when applying it.

1.3 Compensating the discount clock

Fix K, ν, so v is C^2 up to the boundary. Use base controls $A_t \in \mathcal{A}_\nu$, $c_t \in [0, b_K]$, and put

$$\phi_t = \int_0^t c_s ds, \qquad q_t = c_t^2/(2K) \le C_*.$$

For the perturbed process with clock $\theta_t(s)$ choose the discount

$$c_t(s) = c_t/\theta_t(s). \tag{22}$$

This is admissible for the larger cap because $\theta > 1/2$. The covariance rotations are admissible by unitary invariance. The identities

$$\theta_t(s)c_t(s) = c_t, \qquad \theta_t(s)\left(c_t(s)g(X_t^s) - \frac{c_t(s)^2}{2K}\right) = c_t g(X_t^s) - \frac{q_t}{\theta_t(s)} \tag{23}$$

show that the killing clock is exactly independent of s.

For the common localized stopping time γ_s, set

$$J_s = \mathbb{E}\left[e^{-\phi_{\gamma_s}} p_{\gamma_s}(s) v(X_{\gamma_s}^s) + \int_0^{\gamma_s} e^{-\phi_t} p_t(s)\left(c_t g(X_t^s) - \frac{q_t}{\theta_t(s)}\right) dt\right]. \tag{24}$$

Then $J_s \le v(x + s\xi)$. Here is a direct verification, avoiding any filtration issue in the transformed control. In Itô's formula for $e^{-\phi_t} p_t(s) v(X_t^s)$, the Girsanov drift in X^s cancels the quadratic-covariation term with $p(s)$. After adding the integral in (24), the remaining drift divided by $e^{-\phi_t} p_t(s) > 0$ is

$$\theta_t(s)\left[L_{A_t(s)} v - c_t(s) v + c_t(s) g - \frac{c_t(s)^2}{2K}\right] \le 0,$$

where $A_t(s)$ is the admissible unitary conjugate of A_t. This is precisely the larger-cap Bellman inequality. The bounded localizations make the stochastic integrals true martingales.

At $s = 0$, the supremum of J_0 over smaller-cap base controls is $v(x)$, by the usual Bellman verification/dynamic-programming equality and the coincidence of the two value functions. This remains true for the common control-dependent bounded stopping time; an approximately optimal policy for the smaller-cap equation may be stopped at that time. Consequently

$$-\frac{v(x+s\xi) - 2v(x) + v(x-s\xi)}{s^2} \le \sup_{\text{base controls}} \frac{2J_0 - J_s - J_{-s}}{s^2}. \tag{25}$$

For each base control the same γ_s is used in all three terms. There is therefore no differentiation of an indicator of an obstacle-stopping region. The value function being differentiated is the smooth approximant.

1.4 Every changed second-variation term is uniformly bounded

First take $s \to 0$ with the time and quasiderivative localizations fixed. The finite-time convergence in [3, Lemmas 6.2–6.3] justifies the state and density expansions uniformly over the base controls. For each fixed K the cap b_K is finite. Remainder convergence may depend on K, ν, while the estimate obtained after taking this limit will not.

By (13) and the constant killing clock, the terminal second variation is

$$e^{-\phi_\gamma}\left[v_{\xi_\gamma\xi_\gamma}(X_\gamma) + v_{\eta_\gamma}(X_\gamma) + 2Z_\gamma v_{\xi_\gamma}(X_\gamma)\right]. \tag{26}$$

The last two terms have expected absolute value at most $CB(x,\xi)$ by (9), (15), (19), and Cauchy–Schwarz.

The running second variation is exactly

$$\begin{aligned} R_\gamma = \int_0^\gamma e^{-\phi_t}\big[c_t\{g_{\xi_t\xi_t}(X_t) + g_{\eta_t}(X_t) + 2Z_t g_{\xi_t}(X_t)\} \\ - q_t\{8r_t^2 - 2\widehat{r}_t - 4r_t Z_t\}\big]\,dt. \end{aligned} \tag{27}$$

Indeed $(1/\theta)'(0) = -2r$ and $(1/\theta)''(0) = 8r^2 - 2\widehat{r}$, while $p'(0) = Z$ and $p''(0) = 0$. This accounts for all terms introduced by the quadratic penalty.

For the terms with c_t, use the pathwise identity

$$\int_0^\gamma c_t e^{-\phi_t}dt \le 1.$$

Their expected absolute value is at most

$$C\|g\|_{C^2}\mathbb{E}\left[\sup|\xi_t|^2 + \sup|\eta_t| + 2\sup|Z_t\xi_t|\right] \le CB(x,\xi).$$

For the other terms, use $q_t \le C_*$, (15), and (18). Thus

$$\mathbb{E}|R_\gamma| \le CB(x,\xi), \tag{28}$$

uniformly in K, ν and every localization.

For clarity, the passage through the localizations uses the following precise facts. Keep δ, K, ν fixed. Stop at time T and when $|\xi_t| + |Z_t|$ first exceeds N. Then $r, \widehat{r}, P, \pi, \widehat{\pi}$ are bounded, and the finite-time difference-quotient lemmas apply to the common stopping time. After $s \to 0$, [3, Lemma 6.4] controls the difference between the base exit time and the minimum of the three exit times. Formula (16) makes $|\xi_\gamma|^2$ uniformly integrable over controls and stopped times. The probability of the N-cutoff is at most CB/N^2 by (15), while $\mathbb{P}(\tau_\delta > T) \le \|\psi\|_\infty/T$ by (3). Hölder's inequality therefore removes these exceptional events from the terminal Hessian term, bounded for the fixed approximant by $\|D^2 v\|_\infty|\xi_\gamma|^2$. On the remaining events uniform continuity of $D^2 v$ on the fixed closed set $\overline{D}_\delta$, together with the exit-time comparison, gives (21). The intermediate continuity moduli and higher-moment constants may depend on δ, K, ν; their sole role is to make errors tend to zero before these parameters vary. The lower-order terminal and running terms need no convergence statement: their absolute expectations are bounded by CB at every localization, so their bounds survive a limsup.

Combining (25)–(28) in this order yields

$$-v_{\xi\xi}(x) \le CB(x,\xi) + \limsup \sup_{\text{controls}} \mathbb{E}[-e^{-\phi_\gamma} v_{\xi_\gamma\xi_\gamma}(X_\gamma)]. \tag{29}$$

The remaining term is the unchanged terminal term in (21).

1.5 Boundary passage and all real second derivatives

For each fixed K, ν, $v \in C^2(\overline{\Omega})$. If $y \in \partial\Omega$ and ζ is real tangent to the boundary, differentiating $v = g$ along a boundary curve gives

$$|v_{\zeta\zeta}(y)| \le C\left(\|g\|_{C^2} + \|\nabla v\|_{L^\infty(\partial\Omega)}\right)|\zeta|^2 \le C|\zeta|^2. \tag{30}$$

This includes the second fundamental form term; Euclidean tangential second derivatives are not identified with those of an arbitrary extension of g.

By (20), every limiting nontangential direction has a denominator tending to infinity. Limiting tangent directions have numerator bounded by (30). For fixed K, ν, continuity of $D^2 v$ makes the normal and mixed contributions vanish as the directions approach a tangent direction. Normalize $|\zeta| = 1$ and use compactness to obtain

$$\limsup_{\delta \downarrow 0} \sup_{\psi(y)=\delta,\, \zeta \neq 0} \frac{|v_{\zeta\zeta}(y)|}{B_1(y,\zeta)} \le C, \tag{31}$$

where C is independent of K, ν. The order matters: reach the boundary for each fixed smooth approximant before varying the approximation parameters. Its possibly large normal Hessian disappears from this ratio.

Equations (21), (29), and (31) give

$$v_{\xi\xi}(x) \ge -CB(x,\xi). \tag{32}$$

On each compact $E \Subset \Omega$, this is a uniform lower bound $D^2 v \ge -C_E I$. Moreover (6) implies $F_\nu(D^2 v) \le 0$. Since $I/n \in \mathcal{A}_\nu$, $\Delta_{\mathbb{R}^{2n}} v \le 0$. The lower eigenvalue bound and this nonpositive trace bound each real Hessian eigenvalue above by $(2n-1)C_E$. We have proved

$$\|v_{K,\nu}\|_{C^{1,1}(E)} \le C_E \tag{33}$$

with constants depending on Ω, ρ, m, n, E and a C^2 bound for h, but not on K, ν.

1.6 Identification of the limit with the exact envelope

Choose smooth $h_j \to h$ in $C^2(\overline{\Omega})$, $K_j \to \infty$, $\nu_j \to 0$, and put $u_j = -v_{K_j,\nu_j}$ with $g_j = -h_j$. The preceding uniform estimates give a subsequence converging uniformly on $\overline{\Omega}$ and in C^1 on compact subsets to

$$u \in C^{0,1}(\overline{\Omega}) \cap C^{1,1}_{\mathrm{loc}}(\Omega).$$

By (6), $u_j \le h_j + C/\sqrt{K_j}$, so $u \le h$. For each $A \in \mathcal{A}$ set $A_j = (1-\nu_j)A + \nu_j I/n$. Formula (6) gives $L_{A_j} u_j \ge 0$. Passing to distributions shows $L_A u \ge 0$ for all A, hence u is m-subharmonic. The boundary barriers give $u = h$ on $\partial\Omega$.

Let $w \in \mathrm{SH}_m(\Omega)$ with $w \le h$ be any competitor. A fixed sufficiently large C makes $h + C\rho$ m-subharmonic, by the uniform strictness of ρ. Replacing w by $\max(w, h + C\rho)$ increases it and allows us to assume it is bounded. Put $\varepsilon_j = \|h_j - h\|_\infty$. For $t > 0$ the function

$$w_{j,t} = w - \varepsilon_j + t\rho$$

is strictly subharmonic for every L_{A_j}, is at most h_j, and has boundary upper limits at most h_j. If $w_{j,t} - u_j$ had a positive interior maximum, the smooth function u_j plus a constant would touch $w_{j,t}$ from above there. The linear viscosity inequalities would imply $L_{A_j} u_j \ge t\delta_0$ for every A_j, hence $F_{\nu_j}(D^2 v_j) \le -t\delta_0$. On the other hand, at this point $u_j < w_{j,t} \le h_j$, so the penalty vanishes and (6) gives $F_{\nu_j}(D^2 v_j) = 0$, a contradiction. Thus $w_{j,t} \le u_j$. Let $t \downarrow 0$ and $j \to \infty$ to obtain $w \le u$.

The function u is an admissible competitor and dominates every admissible competitor. Therefore $u = P_{m,\Omega} h$, and (33) proves the theorem.

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